PathSync – Intelligent Travel Planning & Booking System

Detailed Product Definition

PathSync is an advanced frontend-driven travel planning system designed to simulate real-world intelligent travel platforms.

It focuses on transforming complex trip planning into a structured, guided, and user-friendly experience.

The system acts as a decision-support interface where users are guided step-by-step from planning to booking.

It uses frontend logic and structured data to simulate intelligent recommendations without relying on backend services.

Extended Objective

The system aims to simplify travel decision-making by reducing cognitive load on users.

It provides structured itinerary generation, cost visibility, and guided booking flow.

The goal is to build a UI that feels intelligent, interactive, and production-ready.

It should reflect how modern SaaS platforms guide users through complex workflows.

Module Explanation (Deep)

Trip Planning Interface: This module captures user intent through structured inputs like destination, duration, and budget. It must ensure clarity and ease of use.

Itinerary Generator: Converts user inputs into a structured day-wise travel plan. The UI must clearly present this data using visual hierarchy.

Booking Interface: Allows users to select hotels and transport options through interactive UI components.

Cost Breakdown System: Displays total and categorized expenses in a clear and readable format.

Recommendation Engine: Provides suggestions using frontend logic to simulate intelligent decision-making.

Saved Trips: Stores previous plans and allows users to revisit them easily.

Frontend Engineering Strategy

The system must follow a component-driven architecture with reusable UI elements.

State management should simulate dynamic updates such as itinerary generation and booking selections.

The UI must maintain strong visual hierarchy and consistent layout.

Interaction flow should be smooth and intuitive, reflecting real-world product standards.

Expected Deliverable (Final)

A production-grade frontend application with structured and scalable UI.

A system that simulates real travel platforms and intelligent planning.

Clean, maintainable, and modular codebase.

A product-ready interface capable of backend integration.